

## Eutric Cambisols

### Description:

Eutric Cambisols are moderately developed mineral soils with a weakly developed cambic (Bw) horizon and relatively high base saturation (>50%). The term “**Eutric**” denotes their richness in basic cations ( $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{K}^+$ ,  $\text{Na}^+$ ), which makes them more fertile and less acidic than **Dystric Cambisols**.

These soils typically form from base-rich parent materials such as basalt, limestone, or volcanic ash under subhumid to semi-humid climates. Their moderate development and balanced nutrient profile make them versatile and agriculturally important in tropical and subtropical regions.

### Drainage:

- **Wet season:** Generally well-drained; occasional short-term water stagnation in depressions or compacted areas.
- **Dry season:** Retains moderate moisture due to balanced texture and humus; rarely waterlogged.
- **Landscape:** Commonly found on gently sloping uplands, footslopes, and terraces derived from basic rocks or alluvial deposits.

### Depth:

- Moderately deep to deep (80–150 cm).
- Rooting depth is usually unrestricted except where stones or compact layers occur.

### Workability:

- **When wet:** Friable and easy to till; does not become excessively sticky.
- **When dry:** Slightly hard but not compacted; easy to cultivate using standard implements.
- Excellent structure and moderate clay content provide good tilth and resistance to crusting.

### **Nutrient Richness & Cation Exchange Capacity (CEC):**

- **CEC:** Moderate to high (15–30 cmol(+)/kg).
- **Base saturation:** High (>50%), indicating high fertility and low acidity.
- **Nutrients:** Nitrogen and phosphorus may become limiting under continuous cropping; potassium and calcium are usually adequate.
- Organic matter content is moderate (1.5–3.0%) and declines under continuous cultivation.
- Micronutrients (Zn, Mn, B) are generally adequate but may need monitoring in intensive systems.

### **pH:**

Slightly acidic to neutral (6.0–7.2), depending on parent material.

### **Management:**

- Maintain soil organic matter through compost, crop residues, or green manures to sustain fertility and structure.
- Apply **balanced N and P fertilizers**; consider phosphorus management to counter mild fixation.
- Use **contour farming or terracing** on slopes to reduce erosion.
- Avoid over-cultivation or heavy machinery, especially on moist soils, to preserve structure.
- Suitable for a wide range of crops including **maize, beans, wheat, coffee, tea, vegetables, and fruit trees**.
- Under irrigation, supports **horticultural crops** such as onions, tomatoes, and cabbages.